

A distributed secondary voltage and frequency controller considering packet dropouts and communication delay

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ABSTRACT

In this paper, the consensus problem is developed for AC Microgrids (MGs) considering communication delay and packet dropouts. The existing control schemes commonly design the secondary control layer assuming ideal communication channels between distributed generation (DG) units. However, in practical situations, the communication links are exposed to packet dropout and delays which can drastically decrease the quality and stability of DGs controllers. Most of the existing literature studies the impact of delay or packet dropout, and the simultaneous effects of both delay and packet dropout in the communication links considering the rigorous stability analysis and statistical properties are proposed in this paper. Therefore, we propose a new distributed control approach to restore the voltage and frequency to the reference values using the collected sample-based technique of each DG unit. In addition, sufficient consensusability conditions are found when the communication network topology, the DG dynamics, the packet dropout rates, and the communication delay are taken into account. The consensus error of the proposed control strategy is determined, and its convergence is shown. Finally, the proposed method is simulated for different scenarios in MATLAB/SimPowerSystems environment to prove the effectiveness of the proposed secondary control scheme.

1. Introduction

The increasing number of RESs and DGs in distribution systems causes unsuitability for utilization of conventional control systems [1]. The concept of MG offers a natural way for these resources to interact while providing various services to the utility. Thus, the MGs help achieve an economical and stable operation of grids, while improving the system flexibility and resiliency [2].

The control scheme of an MG is hierarchical and consists of three control layers, namely, primary, secondary, and tertiary [3]. The first control layer, primary, is based on a decentralized droop control strategy in order to maintain the MG voltage and frequency, and provide power sharing between DGs without requirements for the communication channel. Nevertheless, the steady-state deviations of voltage and frequency from the rated values are occurred in this layer [4]. The conventional secondary control schemes rely on the utilization of an MG central controller, including low bandwidth communication infrastructure to measure some parameters and send back the control output of each DG [5]. The tertiary control scheme exchanges the data with the distribution system operator to optimize the MG operation within the main grid [6].

The secondary control layer compensates for the voltage and

frequency errors, produced inside of the MGs by virtual inertias and output virtual impedances of the primary control layer. This concept has been used by high power utility grids for many years for controlling the frequency of large power networks [7]. Moreover, it has been used in MGs to restore voltage and frequency deviations [8].

There have been proposed several studies on distributed secondary and communication-based control systems. In [9], a multi-agent-based control method is suggested to adjust the output active power of DGs in an MG by considering the impact of uncertainty of communication structure on the stability of the system. The distributed control is capable to amend the load sharing and voltage regulation of MGs by communication links [10]. Moreover, a finite-time distributed frequency control strategy accompanied by a distributed active power sharing controller scheme is proposed in [11]. Furthermore, a consensus-based distributed control approach is suggested in [12] to overcome the voltage and frequency deviations. For restoring the frequency and voltage and maintaining the power-sharing, a pinning-based distributed control method in MGs by considering a communication delay environment is presented in [13]. Moreover, a robust distributed control strategy for frequency and voltage restoration is presented in [14] which considers interval uncertainties within the exchanged information.

Practically, distributed control schemes require essential support of communication links that suffer from different constraints [12]. Packet

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Nomenclature

RES	Renewable energy sources,
MG	Microgrid
DG	Distributed generation,
PI	Proportional-integral
CEL	Compressed edge Laplacian
CIM	Compressed incidence matrix
CIIM	Compressed in-incidence matrix
v_{ref}	Reference voltage value
v_{odi}	Provided nominal value of voltage for controller
V_{ni}	Voltage control input

n_{Qi}	Voltage coefficients
Q_i	Reactive power of i^{th} DG
I_n	Unit matrix of size n
$\otimes$	Kronecker product
k_v	Control performance
e_{vi}	Local neighbour tracking error
τ	Communication delay constant
a_{ij}	The elements of adjacency matrix A
p_{ij}	Bernoulli distribution parameter
ω_{ref}	Reference frequency value
k_{ω}, k_p	Control constants.

dropout in communication links and measurements can adversely affect the stability of MGs [15]. On the other hand, communication delay may degrade the performance of the MG controllers [16]. Therefore, it is indispensable to implement a distributed secondary control strategy for MGs that can cope with the communication delay and packet dropout.

Recently, the impacts of communication impairments on MGs have been evaluated in [17]–[19]. In terms of controlling the stability of MGs with communication delay, several controllers are suggested in a few works to improve this constraint. A multi-agent distributed controller is proposed in [17] for radial DC MGs by a delay-dependent robust controller. In [20], the secondary voltage control system for MGs subject to delays in a connected communication topology by using distributed event-triggered control is solved. A distributed weight average-prediction controller to improve the stability of a time-delayed MG system is proposed in [21]. After small-signal modelling of time-delayed MG, a weight average-prediction-based control system is suggested based on the compensation of the system state. In [22], the optimal structural design of the communication network and pinning location for the distributed secondary voltage control in MGs is investigated and based on the dynamic model of MGs equipped with a time delay communication system, the maximum delays under which the MG can maintain stability are obtained with respect to different sets of controller gains. Besides the communication delays, packet dropouts have occurred in networked systems. In recent studies, the stabilization methods under packet dropout's environment have been investigated in [23]–[26]. In [23], the consensus problem for multiagent systems with time delays and packet losses is investigated, and the sampled-data multiagent systems with packet losses and time-delay are considered. An incremental networked predictive control method based on state observers is suggested in [24] for the cooperative output tracking problem and random communication constraints in the feedback channels of agents are considered, and thus the stability of the resulting closed-loop system is independent of them. In [25], a cooperative control strategy is proposed for islanded MG, which categorized the MG into physical and cyber layers. The main designs in these two layers are two event-triggered techniques, where the first one is utilized for improving the voltage and frequency stability of MG considering the packet loss problem, and the other is used to reduce the communication burden in the control process. In [26], a fully distributed cooperative frequency and voltage restoration secondary control system is presented. A distributed consensus Kalman filter-based estimator is suggested to determine the estimation of the output state with the communication packet loss. In [27], a modified stabilization method is proposed considering packet dropouts. The packet dropouts are modelled by a binary Markov chain for the stabilization of the system [28].

Generally, there are two types of packet dropouts, namely, identical, and non-identical. The identical packet dropouts equally occur in all communication links, whereas the non-identical ones are defined as non-equal and different dropouts in different links [29]. Practically, the majority of pack dropouts in MGs are non-identical, and by using

traditional controllers, or considering only identical packet dropouts, the voltage and frequency stability of MGs can be faced with instability.

A distributed secondary frequency and voltage control scheme for islanded MGs under packet dropouts and delay is proposed in this paper. To the best of the authors' knowledge, this paper is the pioneer to investigate the impact of delay and packet dropout, as uncertain communication links, at one comprehensive control scheme. Most of existing literatures study the impact of delay or packet dropout, the simultaneous effects of both delay and packet dropout in the communication links with considering the rigorous stability analysis and statistical properties are proposed in this paper. The main contributions of this paper are as follows: 1) a novel distributed control method is proposed in order to restore voltage and frequency of each DG unit within a MG and to provide proper active power-sharing. 2) The impact of packet dropouts on the proposed method is investigated, and the stored previously sampled data is utilized to overcome the communication impairments. 3) Different scenarios are simulated to prove the effectiveness of the proposed method under identical and non-identical packet dropouts, with or without delay in an AC MG with meshed topology.

This paper is organized as follows: Section II discusses the problem formulation. In Section III, the modelling of inverter-based MGs is explained. The proposed distributed voltage and frequency controllers are defined in Section IV and Section V, respectively. Finally, Section V concludes the paper.

2. Problem formulation and modelling

A group of DGs interacts with each other through communication networks to achieve collective goals. We assume that modelling exchanged information among DGs depends on the directed graph. The directed graph G is a pair consisting of (ν, ε) , in which $\nu = \{\nu_1, \dots, \nu_N\}$ is defined as a finite set of pairs of nodes, and ε is the set of edges.

The edge (ν_i, ν_j) in the set of edges ε indicates that DG ν_j can obtain packets from the DG ν_i , whereas the opposite direction is not possible. The set of neighboring nodes of ν_i is shown by N_i .

The adjacency matrix $A = [a_{ij}] \in \mathbb{R}^{N \times N}$ is defined with a directed graph $\mathcal{G}$ such that $a_{ii} = 0$, and a_{ij} is a positive value if $(\nu_i, \nu_j) \in \varepsilon$. therefore, the communication layer topology can substitute for the MGs adjacency matrix. The neighbours of DG i is presented by $N_i = \{j | (j, i) \in \varepsilon\}$. In the MG which include N DGs, the voltages and frequencies of DGs play the function of followers, while those references are regarded as the leader. The adjacency matrix of the leader is represented by $g = [g_1, g_2, \dots, g_N]^T$. If DG i gets the references, $g_i = 1$, if not $g_i = 0$. The in-degree of DG i is presented by $deg_i = \sum_{j=1}^N a_{ij}$. The Laplacian matrix $\mathcal{L} = [\mathcal{L}_{ij}] \in \mathbb{R}^{N \times N}$ of the graph $\mathcal{G}$ is defined as $\mathcal{L}_{ii} = \sum_{j \neq i} a_{ij}$ and $\mathcal{L}_{ij} = -a_{ij}$ with $i \neq j$. In compact form, it can be written as $\mathcal{L} = D - A$, where $D = \text{diag}\{deg_1, \dots, deg_N\}$. The eigenvalues of $\mathcal{L}$ are represented by $\lambda_i \in \mathbb{C}$, $i = 1, \dots, N$ and $\lambda_1 = 0$, $|\lambda_1| \leq |\lambda_2| \leq \dots \leq |\lambda_N|$.

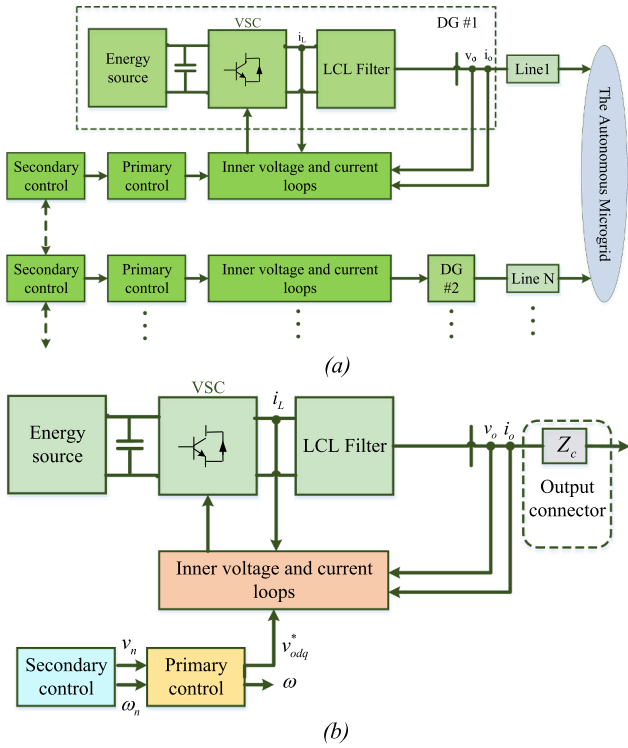


Fig. 1. (a) Single line diagram of a VSC-based MG (b) Inverter-based DG block diagram with its primary and secondary controllers.

In modelling section, a systematic strategy is proposed to model inverter-based MGs. In order to share active power, DG's inverters must be equipped with an external power loop based on the droop scheme. The primary controller consists of a droop controller and two PI controllers in current and voltage loops. In the second layer, a suitable control system is designed to recover the frequency and voltage of AC MGs by considering the two important challenges, i.e., communication delay and packet dropouts.

Fig. 1 (a) shows an inverter-based MG structure operating in islanded mode, control layers, and communication networks. In islanded MGs, DG units are controlled to supply all local loads required power, while regulating voltage and frequency within the permissible ranges.

Fig. 1 (b) depicts the block diagram of the inverter interfaced to an MG through an L or LCL filter. The inverter controller can be categorized into three different loops. The external power control loop, as the first loop, adjusts the voltage amplitude and frequency of the inverter output, based on the droop characteristic. The second and third parts of the control system are voltage and frequency controllers, implemented with the aim of eliminating high-frequency disturbances and providing sufficient attenuation resonance for the LCL filter output [31].

In the context of cooperative control in multi-agent systems, consensusability is a vital and fundamental issue that depends on the control data, and the alignment method. In this paper the agents are DGs, so a consensus is a group of DGs that agree on a common objective of communicating with each other through the communication network or sensors. To solve the consensus control problem, the main task is to design optimally distributed controllers, commonly known as consensus protocols. These protocols mostly operate based on local information from neighbouring DGs.

Initially, the matrix form of MG including N DGs can be represents as $x_i(k+1) = Ax_i(k) + Bu_i(k)$ where $x_i \in \mathbb{R}^n$ is state of DG and $u_i \in \mathbb{R}^m$ is input of DG.

The following assumptions are imposed.

Assumption 1. a. All the eigenvalues of A are either on or outside the unit circle,

b. The pair (A, B) is controllable,

c. The communication graph forms a spanning tree [1].

In the secondary control layer, the controllers communicate with other units through communication networks and using local data and measurements as well as data from adjacent units in order to determine the coordinated behaviours of all units. As mentioned above, communication links between DG units are not always optimal. Moreover, there are a plethora of challenges in the MGs including uncertainty, latency, failure, noise, and packet dropouts. The communication channels are not perfect as typically constrained in practical applications, but are often distorted by packet dropouts and communication delays. The main aim of this paper is to focus on the voltage and frequency control of MGs by considering packet dropouts and communication delays.

3. Proposed distributed voltage control

3.1. Distributed secondary voltage controller during identical packet dropouts

The following assumption is considered for identical packet dropouts.

Assumption 2. At any moment of packet exchange among DGs, all packets pass successfully or fail. This assumption pertains to the cases when DGs are located in a small local area [32].

A distributed voltage controller is designed to regulate the voltage to its reference value. The following decentralized voltage droop control is utilized at the primary stage of the i^{th} DG, as presented in Fig. 1:

$$\begin{aligned} v_{odi} &= V_{ni} - n_{Qi} Q_i \\ v_{odi} &= 0 \end{aligned} \quad (1)$$

The DG output voltage magnitude is:

$$v_{o, \text{magi}} = (v_{odi}^2 + v_{oqi}^2)^{1/2} \quad (2)$$

By utilizing the feedback linearization on (1), the system model becomes.

$$\dot{v}_{odi} = \dot{V}_{ni} - n_{Qi} \dot{Q}_i \equiv u_{vi} \quad (3)$$

Therefore, as a tracking synchronization problem, the restoration voltage of an MG consisting of N DGs is defined by (4).

$$\dot{v}_{od1} = u_{v1}, \dot{v}_{od2} = u_{v2}, \dots, \dot{v}_{odN} = u_{vN} \quad (4)$$

To attain v_{odi} synchronization, it is assumed that DGs communicate with each other via the directed graph. The auxiliary controller u_{vi} is selected based on the data of the DG and its neighbours in the graph as:

$$u_{vi} = -k_v e_{vi} \quad (5)$$

The packet dropouts and network delay are introduced as follows:

$$e_{vi} = \sum_{j \in N_i} a_{ij} \xi_{ij}(k) (v_{odi}(k - \tau) - v_{odj}(k - \tau)) + g_i \xi_{ij}(k) (v_{odi}(k - \tau) - v_{ref}) \quad (6)$$

where the packet dropout denoted by $\xi_{ij}(k)$. $\xi_{ij}(k)$ is a Bernoulli random variable and the pinning gain, $g_i \geq 0$, is the weight of the connected edge from the i^{th} DG to the source. Any alterations in the communication network affect the matrix A . Therefore, the a_{ij} coefficients must change according to the variations in the communication link. For instance, assume that DG_{N+1} is added to the MG, and as a neighbour to the i^{th} DG. Also assume that this new DG sends the packets through an edge with a weight coefficient of a_{iN+1} . Thus, the new term, $a_{iN+1}(v_{odi} - v_{odN+1})$ must be added to the i^{th} local neighbour tracking error, while the tracking errors of the other DGs remain intact.

Assumption 3. It is assumed that $\xi_{ij}(k) = \xi(k)$, where the rate of the packet dropouts of $\xi(k)$ is p , and its average and variance are respectively

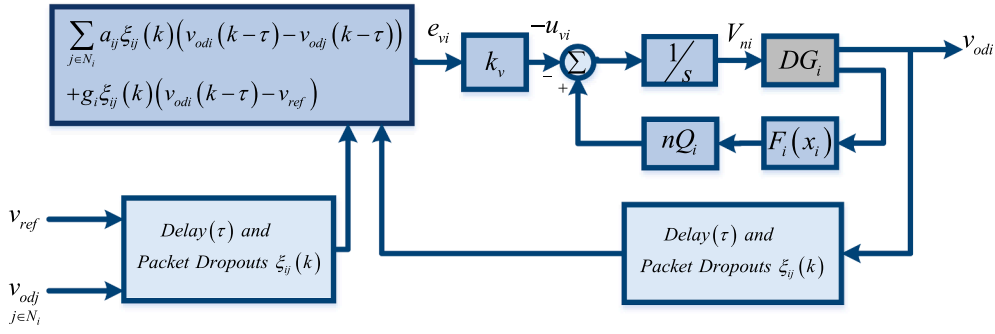


Fig. 2. The secondary voltage control block diagram in case of identical packet dropouts.

$$\mu = 1 - p \text{ and } \sigma^2 = p(1 - p).$$

The probability function of the Bernoulli process is given by:

$$P(\xi_{ij}(k) = 0) = p_{ij}, \quad P(\xi_{ij}(k) = 1) = 1 - p_{ij}, \quad 0 \leq p_{ij} \leq 1 \quad (7)$$

Bernoulli distribution parameter represents the rate of packet dropouts. This variable is determined based on MG reliability. When the packet dropouts are identical, the expected value for Bernoulli's random variable is p .

When the packet transfer between the i^{th} and the j^{th} DGs is successful, $p_{ij} = 0$, otherwise, $p_{ij} \neq 0$. Therefore, the result of Bernoulli's investigation is always either success or failure [33]. In this case, the voltage based on the Bernoulli distribution is obtained as follows:

$$v(k - \tau) = p_{ij}v(k - \tau) + (1 - p_{ij})v(k - \tau - 1) \quad (8)$$

To model the packet dropouts, the values of frequency, voltage, and active power can be determined by using the Bernoulli distribution, and if the $p_{ij} = 1$, the information will be transferred successfully, and otherwise, $p_{ij} = 0$, the proposed distribution controller uses the values of voltage, frequency, and active power from the previously-stored sample.

Theorem 1. The protocol.

$$V_{ni} = \int (u_{vi} + n_{Qi}F_i(x_i))dt, \quad i = 1, \dots, N$$

$$u_{vi} = -k_v \left(\sum_{j \in N_i} a_{ij} \xi_{ij}(k) (v_{odi}(k - \tau) - v_{odj}(k - \tau)) + g_i \xi_{ij}(k) (v_{odi}(k - \tau) - v_{ref}) \right) \quad (9)$$

where, u_{vi} synchronizes the voltage of DG, v_{odi} , with v_{ref} considering packet dropouts and communication delay, as depicted in Fig. 2. Furthermore, we have that.

$$F_i(x_i) \equiv -\omega_c Q_i + \omega_c (v_{oqi} - v_{odi}) \quad (10)$$

Proof: The proof is given below.

Lemma 1. Inspired by [34], in the case of Assumption 1, there are some $P > 0$ such that $P > (1 - \gamma)A^T P A$ if and only if $\gamma > \gamma_c$. In this modified Riccati, $\gamma_c \in [0, 1]$ is the critical value and $\gamma_c = 1 - 1/\rho(A)^2$, $\rho(A) = \max_i |\lambda_i|$. $\rho(A)$ is the spectral radius.

Lemma 2. The stochastic system.

$$x(k + 1) = Ax(k) + \xi(k)Kx(k - \tau) \quad (12)$$

Where $\tau \geq 0$ and the function of the Bernoulli process $\xi(k)$ has mean μ and variance σ^2 according to assumption 3, is mean square stable if there exist $P > 0$ and K such that.

$$(A + \mu K)^T P (A + \mu K) + (\tau \mu^2 + \sigma^2) K^T P K + \tau(A + \mu K - I)^T P (A + \mu K - I) < P \quad (13)$$

Since this Lemma follows nearly the same arguments as Theorem 2.1

in [35], that is omitted in our proof. To prove the theorem1 according to Assumption 1 and lemma 2 the stochastic system with identical packet dropouts and delay is consensusable under the proposed protocol (9) provided that.

$$\eta \triangleq \frac{(\tau + 1)p'}{2\tau p' + 1} [1 - \min_{i \in \mathbb{N}} |\omega \lambda_i - 1|^2], \quad \tilde{\gamma}_c < \eta \quad (14)$$

As to ensure that for any $\gamma > \tilde{\gamma}_c$ in the modified Riccati inequality $P > (1 - \gamma)\tilde{A}^T P \tilde{A}$, it is possible to obtain a positive definite solution P with $\tilde{A} = \tau(A - I) + A$. As a result, the control gain is $K = \gamma \tilde{A}$. which $\gamma = \omega^* (2\tau p' + 1)^{-1} p' = 1 - p, \eta = \gamma$, $\omega^* = \arg \min_{i \in \mathbb{N}} |\omega \lambda_i - 1|^2$ and $\tilde{\gamma}_c$ is the critical value.

prof. Let $\eta = \frac{\omega^*}{2\tau p' + 1} = \frac{\omega^*}{\mu\theta + \sigma^2\mu^{-1}}$. Therefore,

$$\begin{aligned} -\eta &= -\frac{\psi p'}{2\tau p' + 1} \left(1 - \min_{i \in \mathbb{N}} \max_{i=2, \dots, N} |\omega \lambda_i - 1|^2 \right) \\ &= -\frac{\mu \psi}{\mu\theta + \sigma^2\mu^{-1}} \left(1 - \max_{i=2, \dots, N} |\omega^* \lambda_i - 1|^2 \right) \\ &= \frac{\mu \psi}{\mu\theta + \sigma^2\mu^{-1}} \left(\max_{i=2, \dots, N} |(\mu\theta + \sigma^2\mu^{-1})\mu^{-1}\lambda_i - 1|^2 - 1 \right) \\ &= \max_{i=2, \dots, N} \gamma^2 (\sigma^2 + \mu^2\theta) \psi \lambda_i \lambda_i^* - 2\mu\gamma\psi R(\lambda_i) \end{aligned}$$

Where, $\psi = \tau + 1$, $\theta = 2\tau + 1$ and R is real part of λ_i . Furthermore, because $(A|B)$ is controllable, so $(\tilde{A}|B)$ is controllable. The control gain is intended for the delay system with packet dropout $K = \gamma \tilde{A}$ is considered. According to Lemma 1, there are some P that.

$$\begin{aligned} \tilde{A}^T P \tilde{A} + \psi(\sigma^2 + \mu^2\theta) \gamma^2 \lambda_i \lambda_i^* \tilde{A}^T P \tilde{A} \\ - 2\psi\mu\gamma R(\lambda_i) \tilde{A}^T P \tilde{A} \leq \\ \tilde{A}^T P \tilde{A} + \max_{i=2, \dots, N} \psi(\sigma^2 + \mu^2\theta) \gamma^2 \lambda_i \lambda_i^* \tilde{A}^T P \tilde{A} \\ - 2\psi\mu\gamma R(\lambda_i) \tilde{A}^T P \tilde{A} < P \end{aligned}$$

By substituting K , the following inequality are

$$\begin{aligned} \psi^2 (A - \lambda_i K - \tau \psi^{-1} I)^T P (A - \mu \lambda_i K - \tau \psi^{-1} I) \\ + \psi(\sigma^2 + \tau \mu^2) \lambda_i \lambda_i^* K^T P K = \\ (\tilde{A} - \mu \psi \lambda_i K)^T P (\tilde{A} - \mu \psi \lambda_i K) \\ + \psi(\sigma^2 + \tau \mu^2) \lambda_i \lambda_i^* K^T P K \\ = \tilde{A}^T P \tilde{A} + (\psi(\sigma^2 + \theta \mu^2) \lambda_i \lambda_i^* \gamma^2 - \mu \psi \lambda_i \gamma - \mu \psi \lambda_i^* \gamma) \tilde{A}^T P \tilde{A} \\ = \tilde{A}^T P \tilde{A} + (\gamma^2 (\sigma^2 + \theta \mu^2) \lambda_i \lambda_i^* - 2\gamma \mu \psi R(\lambda_i)) \tilde{A}^T P \tilde{A} \\ \leq \tilde{A}^T P \tilde{A} + \left(\max_{i=2, \dots, N} \gamma^2 (\sigma^2 + \theta \mu^2) \lambda_i \lambda_i^* - 2\gamma \mu \psi R(\lambda_i) \right) \tilde{A}^T P \tilde{A} \\ < P \end{aligned}$$

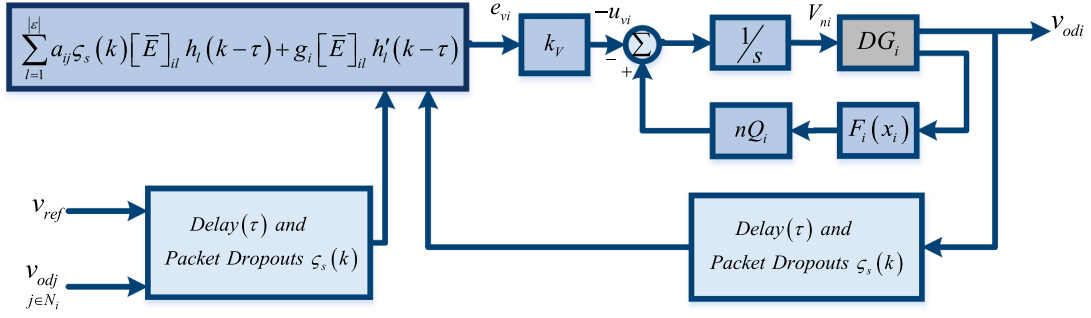


Fig. 3. The secondary voltage control block diagram in case of non-identical packet dropouts.

That is also equivalent to

$$\begin{aligned} & \psi(A - \mu\lambda_i K - \tau\psi^{-1}I)^T P(A - \mu\lambda_i K - \tau\psi^{-1}I) \\ & - \frac{1}{\psi}P + (\sigma^2 + \tau\mu^2)\lambda_i\lambda_i^* K^T P K \\ & = (A - \mu\lambda_i K)^T P(A - \mu\lambda_i K) + (\sigma^2 + \tau\mu^2)\lambda_i\lambda_i^* K^T P K \\ & + \tau(A - \mu\lambda_i K - I)^T P(A - \lambda_i K - I) - P < 0 \end{aligned}$$

Consequently, the system with packet dropout and delay achieves mean square stability synchronously with K , as a result of Lemma 2, so this system with K obtained is mean square consensus.

Distributed control rules are commonly referred to as consensus protocols. In fact, the objective of this paper is to solve the consensusability problem for the DGs with the aim of designing the rules of distributed control, which only utilizes the local data. To this end, the consensus error for the designed control protocol is calculated and its convergence is investigated. Defining the global error vector $e = [e_1^T, \dots, e_N^T]$, as,

$$e = (L \otimes I_n) \begin{bmatrix} \xi(k)v_{odi} - v_{ref} \\ \vdots \\ \xi(k)v_{odN} - v_{ref} \end{bmatrix} \quad (15)$$

Superscript T shows the transpose of matrix e , $I_n \in R^{n \times n}$ is the identity matrix and $\otimes$ is Kronecker product. substituting for u_{vi} from (9) into (4) yields the error dynamic as.

$$e(k+1) = (I_N \otimes A)e(k) - \xi(k)(L \otimes K)e(k - \tau) \quad (16)$$

In this section it is seen that, consensusability problems are inverted into mean square stabilization problem when the packet dropout rates of different communication channels are identical.

3.2. Distributed secondary voltage controller during non-identical packet dropouts

In this section, communication links are considered to have non-identically packet dropouts. The dynamics of the consensus error is determined according to:

$$\delta(k+1) = (I_N \otimes A)\delta(k) + (L(k) \otimes K)\delta(k - \tau) \quad (17)$$

where, for $i \neq j$, $L_{ij}(k) = L_{ij}\xi_{ij}(k)$, and when $i = j$, $L_{ii}(k) = \sum_{j=1}^N L_{ij}\xi_{ij}(k)$. In this case, $\xi_{ij}(k)$ is not independent from the Laplacian matrix, and thus, the investigation of the dynamic of error is complex. To solve this problem, it is assumed that each two DGs communicate with each other via a communication channel. This is not a restrictive assumption, especially for a directionless communication graph. Thus, according to the concept of CIIM and CIM, and the CEL for the graph in [36], the function related to the packet dropouts is independent from the graph topology.

CEL is defined for the graph according to $L_e = E^T \bar{E}$. In addition to simplification, these new concepts standardize consensus analysis for

both direct and indirect communication graphs in cases where packet dropouts are non-identical.

The state variable and the auxiliary state variable on the edge $e_s = (i, j) \in \bar{e}$ for voltage and frequency are respectively defined as $h_s = v_{odi} - v_{odj}$, $h'_s = v_{odi} - v_{ref}$, $z_s = \omega_i - \omega_j$ and $z'_s = \omega_i - \omega_{ref}$, and $\zeta_s = \xi_{ji}$ denotes the packet dropouts on e_s at a rate of p_s . The voltage control protocol in this case is obtained as follows:

$$u_{vi}(k) = -k_v \left(\sum_{l=1}^{|e|} a_{ij}\zeta_s(k)[\bar{E}]_{il}h_l(k - \tau) + g_i\zeta_s(k)[\bar{E}]_{il}h'_l(k - \tau) \right) \quad (18)$$

Assumption 4. It is assumed that For this case, the Bernoulli function $\zeta(k)$ is considered with mean $\Lambda = \text{diag}\{1 - p_1, \dots, 1 - p_{|\bar{e}|}\}$ and the covariance matrix $\Sigma = [\sigma_{ij}] \in \mathbb{R}^{|\bar{e}| \times |\bar{e}|}$ which $\sigma_{ij} = E\{(\zeta_i - p_i + 1)(\zeta_j - p_j + 1)\}$ for $i \neq j$ and $\sigma_{ii} = p_i(1 - p_i)$.

The voltage-state space corresponding to each edge $e_s = (i, j) \in \bar{e}$ is then obtained as follows:

$$h_s(k+1) = Ah_s(k) + u_{vi}(k) - u_{vj}(k) \quad (19)$$

According to the equation $L_e = E^T \bar{E}$, we have.

$$[L_e]_{sl} = \sum_{m=1}^N [E]_{ms}[\bar{E}]_{ml} = [E]_{is}[\bar{E}]_{il} + [E]_{js}[\bar{E}]_{jl} = [\bar{E}]_{il} - [\bar{E}]_{jl} \quad (20)$$

By combining (19) and (20), the voltage-state space of each edge $e_s = (i, j) \in \bar{e}$ is obtained as $h(k) = [h_1^T(k) \dots h_{|\bar{e}|}^T(k)]^T$ and $\zeta(k) = \text{diag}\{\zeta_1(k) \dots \zeta_{|\bar{e}|}(k)\}$.

Moreover,

$$e(k+1) = (I_{|\bar{e}|} \otimes A)e(k) - (L_e \zeta(k) \otimes K)e(k - \tau) \quad (22)$$

The aforementioned results are summarized in the Theorem 2.

Theorem 2: The following control protocol and (24).

$$V_{ni} = \int (u_{vi} + n_{Qi}F_i(x_i))dt, \quad i = 1, \dots, N \quad (23)$$

leads to the synchronization of v_{odi} to the value of v_{ref} , considering the impacts of communication constraint and non-identical packet dropouts in the communication network, as shown in Fig. 3.

Proof: The proof is presented in given below.

At first step, we expand lemma 2 to consider the case of nonidentical packet dropout and delay to infer consensusability conditions.

Lemma 3. The stochastic system.

$$x(k+1) = Ax(k) + \zeta(k)Kx(k - \tau) \quad (25)$$

Where $\tau \geq 0$ and the function of the Bernoulli process is $\zeta(k) = \text{diag}\{\zeta_1(k), \dots, \zeta_n(k)\}$ which has mean $\Lambda = \text{diag}\{p'_1, \dots, p'_{|\bar{e}|}\}$ and covariance $\Sigma = E\{(\zeta_i - p_i + 1)(\zeta_j - p_j + 1)\}$ for $i \neq j$ and $\Sigma = p_i(1 - p_i)$

for $i = j$. according to assumption 4, is mean square stable if there exist $P > 0$ and K such that.

$$(A + \Lambda K)^T P (A + \Lambda K) + \tau K^T \Lambda P \Lambda K + K^T (\Sigma \odot P) K + \tau (A + \Lambda K - I)^T P (A + \Lambda K - I) < P \quad (26)$$

$\odot$ represent the Hadamard product. Now we establish a sufficient condition for consensusability. The solution to which can be found by solving the feasibility problem over real numbers.

$$\text{Let } H(k) = [h_1^T(k), \dots, h_{|\bar{e}|}^T(k)]^T.$$

That is what we have $H(k+1) = (I_{|\bar{e}|} \otimes A)H(k) - (L_e \zeta(k) \otimes K)H(k - \tau)$ (27)

Suppose we have assumption 1, $E = [E_t, E_r]$ and $\bar{E} = [\bar{E}_t, \bar{E}_r]$, which $E_t, \bar{E}_t$ refer to the edges on the spanning tree and $E_r, \bar{E}_r$ refer to the edges on the resting edges. According to [37], we have that.

- It is possible to find a matrix T such that $E_r = E_t T$,
- L_e is like $\begin{bmatrix} MR^T & M\theta \\ 0 & 0_{(|\bar{e}|-N+1) \times (|\bar{e}|-N+1)} \end{bmatrix}$, which the null space of E has an orthonormal basis $\theta.M = E_t^T \bar{E}_r, R = [I, T]$ and MR^T gives the no zero eigenvalues of L_e .

As a consequence, $H(k) = [h_1^T(k), \dots, h_r^T(k)]^T$, which H_t and H_r demonstrate the edge states on the spanning tree and the resting edge states respectively. The following result is based on a similar approach presented in [37].

Lemma 4. if $\mathcal{G}$ has a directed graph, then $H_r = (T^T \otimes I)H_t$.

As a result of Lemma 4,

$$\begin{aligned} & H_t(k+1) \\ &= (I \otimes A)H_t(k) - ([E_t^T \bar{E}_t \zeta_r(k) \otimes K)H_t(k - \tau) \\ & - ([E_t^T \bar{E}_t \zeta_r(k) \otimes K)H_r(k - \tau) = (I \otimes A)H_t(k) \\ & - ([E_t^T \bar{E}_t \zeta_r(k) + [E_t^T \bar{E}_t \zeta_r(k) T^T \otimes K)H_t(k - \tau) \\ & = (I \otimes A)H_t(k) - (M \zeta_r(k) R^T \otimes K)H_t(k - \tau) \quad (28) \end{aligned}$$

By considering lemma 4, since protocol (28) is mean square stable with some K , protocol (27) should be mean square stable with K , which amounts to that the proposed protocol (23) achieves mean square consensus with K .

For the proof of Theorem 2, assume Assumption 1 and apply lemma 3. So, the stochastic system with non-identical packet dropouts and delay is mean square consensusable under the proposed protocol (23) which there are some $\kappa \in \mathbb{R}$ that.

$$\tilde{\gamma}_c < \psi \kappa (M \Lambda R^T + R \Lambda M^T) - \psi \kappa^2 (\theta R \Lambda M^T M \Lambda R^T + R (\Sigma \odot M^T M) R^T)$$

Therefore, based on assumption 1 and lemma 1, there exist some $P > 0$ that.

$$\begin{aligned} I \otimes P &> I \otimes \tilde{A}^T P \tilde{A} - \{ \psi (\kappa (M \Lambda R^T + R \Lambda M^T) - \kappa^2 R (\Sigma \odot M^T M) R^T) \\ &- \kappa^2 \theta \psi R \Lambda M^T M \Lambda R^T \} \otimes \tilde{A}^T P \tilde{A} \end{aligned}$$

Let $K = \kappa \tilde{A}$ and $P = I \otimes P$. As a consequence, the inequality above equals.

$$\begin{aligned} & [I \otimes \tilde{A} - \psi M \Lambda R^T \otimes K]^T P [I \otimes \tilde{A} - \psi M \Lambda R^T \otimes K] \\ & + \tau \psi (M \Lambda R^T \otimes K)^T P M \Lambda R^T \otimes K + \psi (R \otimes K)^T [(\Sigma \otimes 11^T) \\ & \odot M^T P M (R \otimes K)] \\ & < P \end{aligned}$$

Substitute $\tilde{A} = \tau(A - I) + A$ into formal inequality and follow the same steps as.

$$\begin{aligned} & (I \otimes A - M \Lambda R^T \otimes K)^T P (I \otimes A - M \Lambda R^T \otimes K) \\ & + \tau (M \Lambda R^T \otimes K)^T P (M \Lambda R^T \otimes K) \\ & + (R \otimes K)^T [(\Sigma \otimes 11^T) \odot M^T P M] (R \otimes K) \\ & + \tau (I \otimes A - M \Lambda R^T \otimes K - I)^T P (I \otimes A - M \Lambda R^T \otimes K - I) < P \end{aligned}$$

Considering Lemma 3, the system with non-identical packet dropouts and delay is mean square stable with K , indicating that is mean square consensusable with this designed control gain K .

Corollary 1. By considering Assumption 1 and all the eigenvalues of A are either on or outside the unit circle, when $M \Lambda R^T + R \Lambda M^T > 0$ the MG is mean square consensusable under the proposed protocol in theorem 2 if.

$$\tilde{\gamma}_c \leq \frac{\psi \lambda_{\min}^2 M \Lambda R^T + R \Lambda M^T}{4 \left(\theta \rho(RR^T) \rho(\Lambda M^T M \Lambda) + \max_i \sigma_{ii} \rho(M^T M) \rho(RR^T) \right)}$$

When $\Lambda = p' I$, the MG under a protocol (23) is mean square consensusable if.

$$\tilde{\gamma}_c \leq \frac{\psi p' \lambda_{\min}^2 (MR^T + RM^T)}{4 (\theta p' \rho(RR^T) \rho(M^T M) + p \rho(M^T M) \rho(RR^T))}$$

We have $M \Lambda R^T + R \Lambda M^T > 0$, therefore, it is clear that $\lambda_{\min}(M \Lambda R^T + R \Lambda M^T) > 0$.

$$\text{Let } F(\kappa) = \kappa \lambda_{\min}(M \Lambda R^T + R \Lambda M^T) - \kappa^2 \left(\theta \rho(RR^T) \rho(\Lambda M^T M \Lambda) + \max_i \sigma_{ii} \rho(M^T M) \rho(RR^T) \right).$$

Then by calculation $\frac{dF(\kappa)}{d\kappa}$, the maximum value for $F(\kappa)$ is obtained and the value of κ is

$$\kappa = \frac{\psi \lambda_{\min}^2 M \Lambda R^T + R \Lambda M^T}{2 \left(\theta \rho(RR^T) \rho(\Lambda M^T M \Lambda) + \max_i \sigma_{ii} \rho(M^T M) \rho(RR^T) \right)}$$

Thus, the MG under proposed protocol (23) with $K = \kappa \tilde{A}$ achieves to mean square consensusability.

In this section, the dynamic over edges is computed by using the compressed edge Laplacian for digraph to separate the packet dropouts from the network topology when the packet dropouts are not identical.

4. Proposed distributed frequency control

4.1. Distributed secondary control of frequency during non-identical packet dropouts

A distributed controller is designed in this section to restore the frequency ω_i of DG units to the reference value. In order to pick up the proper control inputs for the droop characteristic, i.e., ω_{ni} , secondary frequency control is designed.

The following decentralized frequency droop control is utilized at the primary stage of the i^{th} DG.

$$\omega_i = \omega_{ni} - m_{P_i} P_i \quad (29)$$

Its derivative is:

$$\dot{\omega}_1 = u_{ni} - m_{P_i} \dot{P}_i = u_{oi} \quad (30)$$

where u_{oi} is an auxiliary control signal which is used to linearize the input-output map [26]. Thus, ω_{ni} can be obtained in terms of u_{oi} . Based on (30), the secondary frequency control can be converted to a synchronization tracking problem with first-order equations and linear multi-agent systems.

$$\dot{\omega}_1 = u_{o1}, \dot{\omega}_2 = u_{o2}, \dots, \dot{\omega}_N = u_{oN} \quad (31)$$

It is assumed that DGs communicate with each other through a direct graph to achieve synchronization of ω_{ni} . The auxiliary controller u_{oi} is

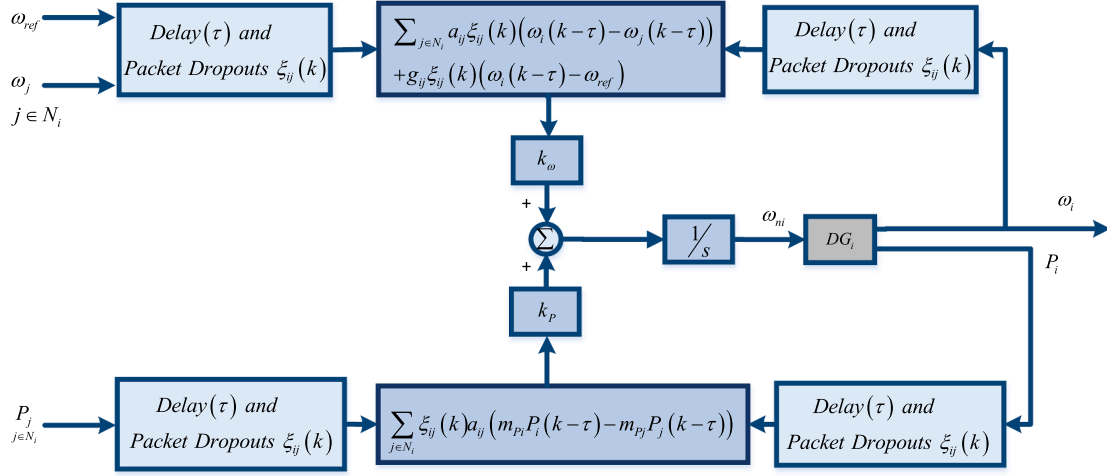


Fig. 4. The secondary frequency control block diagram in case of identical packet dropouts.

selected based on the information of the DG and its neighbour in the graph as follows:

$$u_{\omega i} = -(k_{\omega} e_{\omega i} + k_p e_{p_i}) \quad (32)$$

The local neighbour tracking error including the impacts of delays and packet dropouts is:

$$e_{\omega i} = \sum_{j \in N_i} a_{ij} \xi_{ij}(k) (\omega_i(k-\tau) - \omega_j(k-\tau)) + \xi_{ij}(k) (\omega_i(k-\tau) - \omega_{ref}) \quad (33)$$

It should be noted that the active power between the DGs is shared by the droop controller according to:

$$P_j m_{pj} = P_i m_{pi}, \quad \forall i, j \in N \quad (34)$$

Therefore, the local neighbour tracking error is obtained by:

$$e_{p_i} = \sum_{j \in N_i} \xi_{ij}(k) a_{ij} (m_{pi} P_i(k-\tau) - m_{pj} P_j(k-\tau)) \quad (35)$$

The auxiliary voltage control signal is also obtained from (36) which incorporates the effects of communication delays and packet dropouts. The frequency is obtained based on the Bernoulli distribution as follows:

$$\omega(k-\tau) = p_j \omega(k-\tau) + (1-p_j) \omega(k-\tau-1) \quad (37)$$

Substituting (37) into (33), the vicinity tracking error $e_{\omega i}$ is then obtained from (38).

Furthermore, substituting (37) into (35), the local neighbour tracking error e_{p_i} is calculated from (39).

The block diagram of the distributed frequency-based secondary controller is depicted in Fig. 4. The control input ω_{ni} is defined as follows:

$$\omega_{ni} = \int u_{\omega i}, \quad i = 1, 2, \dots, N \quad (40)$$

The control gains k_{ω} , k_p modify the convergence rate of DG frequencies and the output power of the filter.

Theorem 3: The consensus in (32) leads to synchronization of ω_{ni} to ω_{ref} under the impacts of delays and packet dropouts in the communication link.

Proof: The proof is similar to the proof of theorem 1.

As seen from (32), the protocol $u_{\omega i}$ is made by two terms. The first term guarantees the tracking of the nominal frequency, i.e., the value of ω_{ni} reaches ω_{ref} , whereas the second term guarantees the exact distribution of the active power, i.e., $m_{pi} P_i = m_{pj} P_j$.

4.2. Distributed secondary control of frequency during non-identical packet dropouts

The state variable and the auxiliary state variable on the edge $e_s = (i, j) \in \bar{e}$ for the frequency are respectively defined as $z_s = \omega_i - \omega_j$ and $z'_s = \omega_i - \omega_{ref}$. The packet dropout on e_s at the rate of p_s , is denoted by $\zeta_s = \xi_{ij}$. Therefore, the frequency control protocol in this case is:

$$u_{\omega i}(k) = -k_{\omega} \left(\sum_{l=1}^{|\bar{e}|} a_{ij} \zeta_s(k) [\bar{E}]_{il} z_l(k-\tau) + g_i [\bar{E}]_{il} z'_l(k-\tau) \right) \quad (41)$$

Similar to the section IV, the frequency-state space corresponding to each $e_s = (i, j) \in \bar{e}$ is obtained from (42), where $Z(k) = [z_1^T(k) \dots z_{|\bar{e}|}^T(k)]^T$.

Theorem 4: The following control protocol.

$$\omega_{ni} = \int (u_{\omega i} + n_{pi} S_i(x_i)) dt, \quad i = 1, \dots, N$$

$$u_{\omega i}(k) = -k_{\omega} \left(\sum_{l=1}^{|\bar{e}|} a_{ij} \zeta_s(k) [\bar{E}]_{il} z_l(k-\tau) + g_i [\bar{E}]_{il} z'_l(k-\tau) \right) \quad (43)$$

$$\begin{aligned} \epsilon z_s(k+1) &= A z_s(k) + [u_{\omega i}(k) - u_{\omega j}(k)] = A z_s(k) - k_{\omega} \left(\sum_{l=1}^{|\bar{e}|} \zeta_l(k) \right. \\ &\quad \left. ([\bar{E}]_{il} - [\bar{E}]_{jl}) z_l(k-\tau) + g_i \zeta_i(k) ([\bar{E}]_{il} - [\bar{E}]_{jl}) z'_l(k-\tau) \right) \\ &= A z_s(k) - k_{\omega} \left(\sum_{l=1}^{|\bar{e}|} \zeta_l(k) [L_e]_{sl} z_l(k-\tau) + g_i \zeta_i(k) [L_e]_{sl} z'_l(k-\tau) \right) \end{aligned} \quad (42).$$

undertakes that the DG's output frequency, ω_{ni} synchronizes with ω_{ref} irrespective of communication delays and packet dropouts in communication links. The aforementioned consensus method results in the synchronization of ω_{ni} to ω_{ref} . Furthermore,

$$S_i(x_i) \equiv -\omega_c P_i + \omega_c (v_{odi} i_{odi} + v_{oqi} i_{oqi}) \quad (44)$$

Proof: The proof is similar to the proof of theorem 2.

5. Simulation results

Four different scenarios are simulated in the MATLAB/SimPowerSystems to analysis the effectiveness of the proposed controller protocol for islanded MGs with rated voltage of 380 V and frequency of 50 Hz, as shown in Fig. 5. The islanded MG consists of two loads and four inverter-based DG units. The loads are modelled by series RL branches,

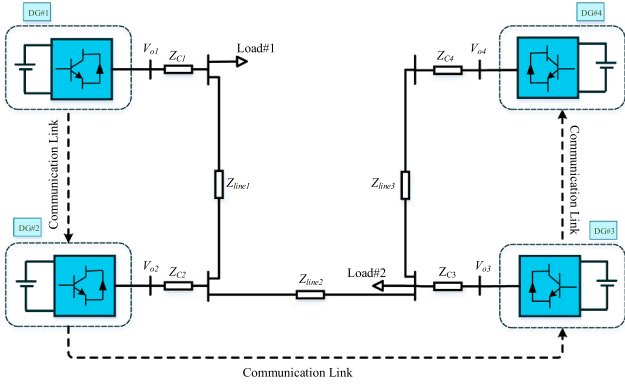


Fig. 5. The case study of islanded MG.

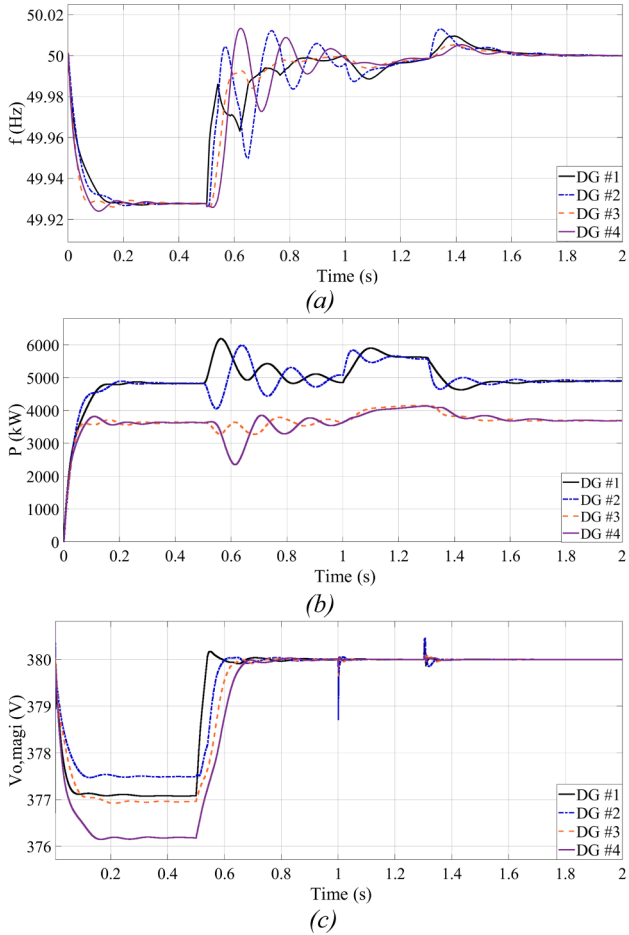


Fig. 6. The results of control scheme in scenario 1: identical packet dropouts for (a) Frequency (b) Active power (c) Voltage.

and the parameters of the study system are detailed in Appendix A. K_{PV} , K_{IV} and K_{PC} , K_{IC} are respectively the voltage and current control parameters.

Scenario 1: Identical packet dropouts.

The consensus protocol function is designed at $t = 0$ s, and only the primary controller (voltage and frequency droop controllers) is in the active mode. As seen from Fig. 6 (a), at $t = 0.5$ s, identical packet dropouts have occurred at the rate of 40% and then the operation of the consensus secondary controller is started. At $t = 1$ s, the load is connected to DG₂, and at $t = 1.3$ s, it is disconnected.

Scenario 2: Non-identical packet dropouts.

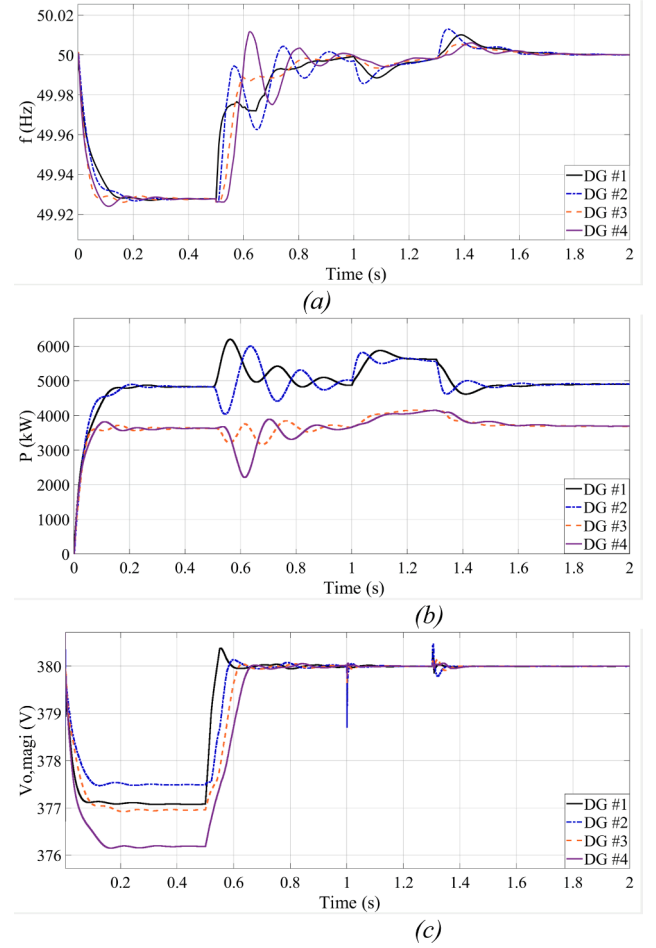


Fig. 7. The results of control scheme in scenario 2 (a) Frequency (b) Active power (c) Voltage.

It is assumed that the first, second, third, and fourth channels undergo 20%, 35%, 40%, and 40% packet dropouts, respectively. During the protocol design for the case of non-identical packet dropouts, at $t = 0$ s, only the primary controller (voltage and frequency droop controllers) is activated. Then, at $t = 0.5$ s, the secondary consensus controller is activated and the packet dropouts occur at the rate of 20% in the first channel, 35% in the second channel, 40% in the third channel, and 40% in the fourth channel, as shown in Fig. 7. Consequently, the consensus-based controllers continue to operate and restore the voltage and frequency of droop control after the packet dropouts occur in both mode of identical and non-identical packet dropouts, as presented in Fig. 6 (a) and (b) and 7 (a) and (b). Moreover, the power sharing among DGs is performed successfully, Fig. 6 (c) and 7 (c).

Scenario 3: Non-identical packet dropouts with delays.

In this scenario, the non-identical packet dropouts occur with a same rate of Scenario 2, that constant delay applying accidentally is considered $\tau = 4$ ms. As observed from Fig. 8, the proposed scheme can restore the reference values of the voltage and frequency when the non-identical packet dropouts and the constant delays are imposed.

Scenario 4: Non-identical packet dropouts with non-uniform delays.

The events of this scenario and Scenario 2 are the same, but, in this scenario, the non-identical packet dropouts occur at a same rate of Scenario 2, which a non-uniform delay $2\sin(0.1t)$ s is imposed in communication channels. As shown in Fig. 9, the non-identical packet dropouts and time-varying delays in communication channels are eliminated through the proposed protocol.

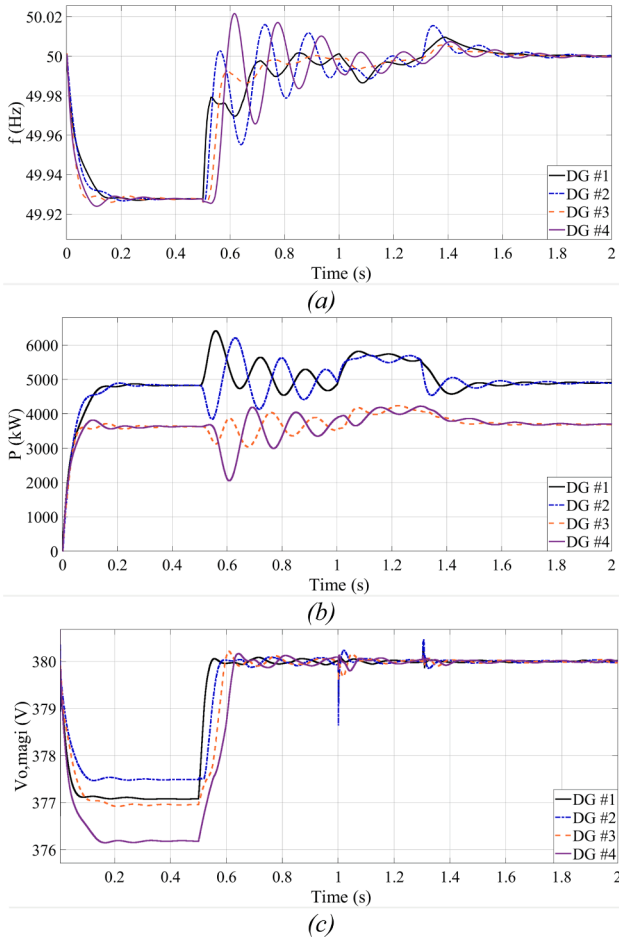


Fig. 8. The results of control scheme in scenario 3 (a) Frequency (b) Active power (c) Voltage.

6. Comparison

The results of the proposed control method are compared with other previously reported studies published in [36]–[40], as shown in Table 1. In [36], a consensus-based distributed voltage and frequency control method is suggested, and the impact of delay is also considered in [36]. Table A1.

However, the restoration time is around 1 s, which demonstrates the low speed of recovery of voltage and frequency by this method. A distributed voltage control is presented in [37] for only voltage control, and the frequency restoration is not considered in [37]. In [38], a power management strategy for MGs is represented, which only restores the frequency of the system to the reference value. An adaptive decentralized droop controller is presented in [39] for only frequency restoration within a low period of time. In all aforementioned methods, the absence of a comprehensive method with consideration of the impact of packet dropout and delay for both voltage and frequency control with high restoration speed is a challenging research gap. In [40], the voltage control by consideration of both packet dropout and delay is presented, however, the frequency control is not considered, and the restoration speed is too low. Table A2.

In the proposed scheme, the impact of packet dropout and delay for both voltage and frequency control with high restoration speed is considered, and the proposed method restores the voltage and frequency to the reference value very fast under communication constraints.

We have tried to obtain simulation results considering the practical conditions as much as possible. In this regard, the following practical issues have been taken into account:

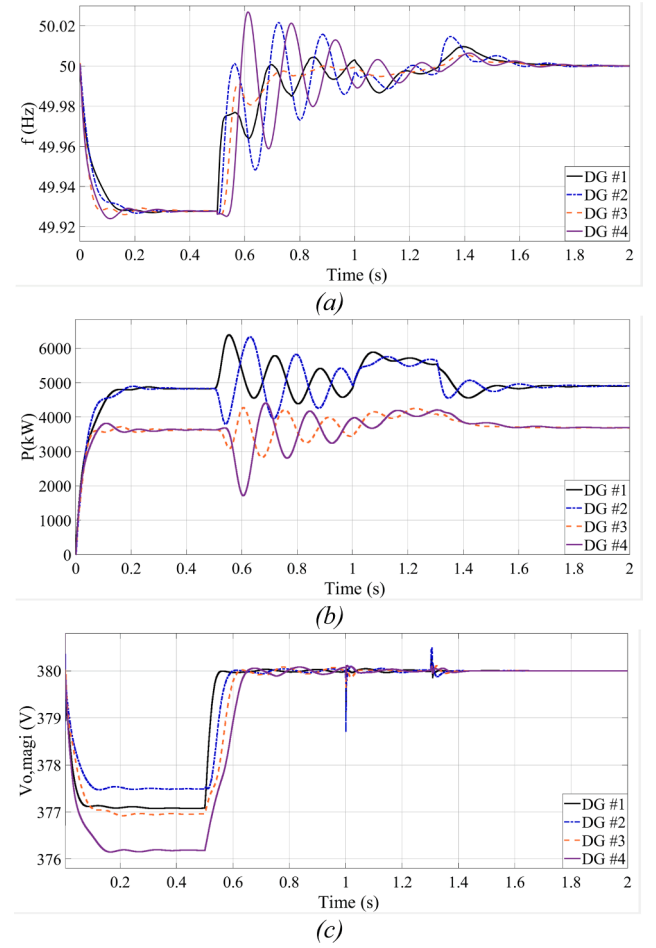


Fig. 9. The results of control scheme in scenario 4 (a) Frequency (b) Active power (c) Voltage.

Table 1

Parameters of the Microgrid test system.

DG 1 and 2 (9 kVA rating)		DG 3 and 4 (7 kVA rating)	
Parameter	Rating	Parameter	Rating
m_p	9.4×10^{-5}	m_p	12.5×10^{-5}
n_Q	1.3×10^{-3}	n_Q	1.4×10^{-3}
R_c	0.03Ω	R_c	0.03Ω
L_c	0.35 mH	L_c	0.35 mH
R_f	0.1Ω	R_f	0.1Ω
L_f	1.35 mH	L_f	1.35 mH
C_f	$50 \mu\text{F}$	C_f	$50 \mu\text{F}$
K_{PV}	0.1	K_{PV}	0.05
K_{IV}	420	K_{IV}	390
K_{PC}	15	K_{PC}	10.5
K_{IC}	20,000	K_{IC}	16,000
c	31.41		

- The intrinsic communication link delays have been taken into account in the simulation scenarios.
- The controllers for each DG unit are digitally implemented (suitable for using in DSP or a microcontroller) in the MATLAB/SimPowerSystems software.
- The detailed switching models are used for inverters.
- A time delay has been imposed on the output of PWM controller before being applied to the switches.
- The signal sampling delay has been taken into account before processing in the inverter's control loop.

Table A1

Comparison of the proposed control scheme with other existing methods.

Method	Voltage control	Frequency control	Active power sharing	Consensus	Packet dropout consideration	Delay consideration	Restoration time
[33]	Yes	Yes	Yes	Yes	No	Yes	1 s
[34]	Yes	No	Yes	Yes	No	Yes	0.200 s
[35]	No	Yes	No	No	No	No	–
[36]	No	Yes	No	No	No	No	0.150 s
[37]	Yes	No	Yes	Yes	Yes	Yes	3 s
Proposed method	Yes	Yes	Yes	Yes	Yes	Yes	0.5 s

Table A2

Parameters of the line of test system.

Line 1		Line 2	
Parameter	Rating	Parameter	Rating
R _{l1}	0.23 Ω	R _{l2}	0.35 Ω
L _{l1}	318 μ H	L _{l2}	1847 μ H
Line 3			
Parameter	Rating		
R _{l3}	0.23 Ω		
L _{l3}	318 μ H		

Four different scenarios are implemented for both identical packet and non-identical packet dropouts and also constant and non-uniform delays are considered.

7. Conclusion

In this paper, a new distributed secondary control strategy is proposed for converter-based MGs by considering the packet dropouts and communication delay. The frequency and voltage restoration of DGs, as well as the optimal active power-sharing of MG are achieved. The collected sample-based technique is employed to avoid the impact of communication constraints among DGs, which can significantly improve the stability of MGs. Finally, the performance of the proposed scheme is validated by simulations performed in MATLAB/SimPowerSystems that shows the consensusability and effectiveness of the proposed strategy. To further improvements, the application of the proposed method can be expanded on hybrid AC/DC and DC MGs. Furthermore, the existence of cyber-attacks on communication links can be considered to enhance the reliability of communication links.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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